

NIVEDITHA K

AI Engineer

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SUMMARY

I am an AI Engineer with over 3 years of experience specializing in the development and deployment of advanced Machine Learning, Deep Learning, and NLP solutions. My expertise includes designing scalable end-to-end pipelines, utilizing large language models, and building systems that provide actionable insights from complex data. I excel in creating innovative AI applications that enhance business processes and improve decision-making

EXPERIENCE

08/2023 - Present

Chennai Tamilnadu, India

- AI Engineer
Spiro Prime Tech Services
Tech company specializing in AI and machine learning solutions
 - Designed and Build machine learning and deep learning models (ANN, CNN, RNN) for classification, prediction, and pattern recognition using TensorFlow
 - Developed end-to-end Natural Language Processing (NLP) pipelines using NLTK and Hugging Face Transformers for sentiment analysis, text classification, and automatic summarization
 - Built Retrieval-Augmented Generation (RAG) systems using Gemini AI, Chroma DB, and Lang Chain for document question answering

EDUCATION

06/2020 - 05/2023

Thanjavur, Tamilnadu, India

- Bachelor of Computer Applications (BCA)
Bon Secours College for Women

05/2018 - 05/2020

Mannargudi, Tamilnadu, India

- HSSC (STATE BOARD)
ST Joseph's Girls Higher Secondary School

05/2017 - 05/2018

Pullavarayan Kudi kadu, Tamilnadu, India

- SSLC (STATE BOARD)
Government High School

LANGUAGES

Tamil
Native



English
Proficient



SKILLS

Scikit-learn	Deep Learning	NLP	TensorFlow	NLTK	Sentiment Analysis	Flask	Keras	PyTorch
LSTM	OpenCV	Pandas	Seaborn	Visual Studio Code	Python	SQLite	Django	MLflow
ML Algorithms	SpaCy	Git	REST	Django Rest Framework	Docker	Microservices	React	MFC

PROJECTS

AI-Based Disease Prediction & Specialist Recommendation System

10/2023 - 11/2024

AI-Based Disease Prediction & Specialist Recommendation System

- Trained and benchmarked multiple ML and DL classification models; built ANN using TensorFlow achieving high accuracy for multi-disease classification across 15+ disease categories
- Build an NLP-based symptom analysis system using tokenization, TF-IDF/embeddings, and classification algorithms to predict diseases from user-input symptoms
- Integrated an intelligent doctor recommendation system that suggests relevant medical specialists based on predicted disease categories, enhancing healthcare decision support
- Engineered RESTful APIs using Django REST Framework for real-time model inference, integrated with frontend applications
- Applied SMOTE and class-balancing techniques to handle imbalanced medical datasets; improved model F1-score by optimizing feature engineering pipeline

PROJECTS

Smart Help - AI-Driven HR Management System

📅 01/2025 - 03/2026

Smart Help - AI-Driven HR Management System (RAG + Multi-Agent LLMs)

- Designed and built a multi-agent orchestration framework using LangChain, CrewAI, and LangGraph to automate complex HR workflows including onboarding, meeting scheduling, and event planning
- Built custom AI agents (Query Analyzer, Scheduler, Event Planner, Communication Agent) enabling end-to-end task decomposition and autonomous execution of HR processes
- Architected production-grade RAG pipelines integrating LangChain with internal HR document vector stores for context-aware, policy-compliant Q&A with low hallucination rates
- Fine-tuned LLMs using Hugging Face Transformers with custom prompt templates tailored for HR use cases, improving response relevance and accuracy
- Built secure backend microservices using FastAPI and Django REST Framework, deployed and tested in sandboxed enterprise environment using Docker
- Integrated OpenCV-based document preprocessing pipeline for automated employee ID verification and data extraction workflows

Real-Time Ambulance Detection System

📅 01/2022 - 05/2023

Real-Time Ambulance Detection System

- Built and curated a multimodal dataset with 300+ siren recordings and 10K+ traffic images, enhancing model accuracy and real-world generalization
- Engineered a real-time ambulance detection pipeline with low-latency inference, enabling intelligent traffic signal control and faster emergency response